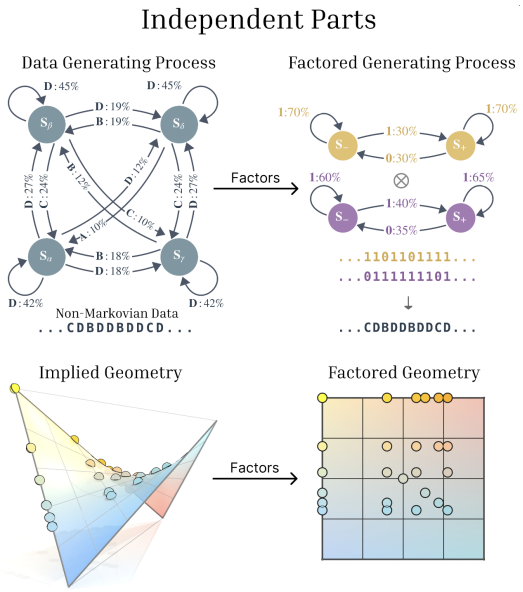

Figure 5. A minimal example (two SNS factors) to compare two candidate representations (joint vs. factored). The joint representation (bottom left) is a curved 2d submanifold of a 3-simplex, and varies in three dimensions. The factored representation (bottom right) contains belief updates over each factor in an orthogonal subspace, and so only varies in two dimensions.

operators acting on the joint $\mathbb{R}^4$ space: $T^{(\mathrm{A})} = T_1^{(0)} \otimes T_2^{(0|0)}$, $T^{(\mathrm{B})} = T_1^{(0)} \otimes T_2^{(1|0)}$, $T^{(\mathrm{C})} = T_1^{(1)} \otimes T_2^{(0|1)}$, and $T^{(\mathrm{D})} = T_1^{(1)} \otimes T_2^{(1|1)}$.

It's now useful to define $T_2^{|z^{(1)}} = \sum_{z^{(2)} \in \mathcal{Z}_2} T_2^{(z^{(2)}|z^{(1)})} = T_2^{(0|z^{(1)})} + T_2^{(1|z^{(1)})}$, which is row-stochastic for each $z^{(1)} \in \mathcal{Z}_1$.

The net row-stochastic transition matrix on the joint latent space $T = \sum_{x \in \{\mathrm{A,B,C,D}\}} T^{(x)} = T_1^{(0)} \otimes T_2^{|0} + T_1^{(1)} \otimes T_2^{|1}$ has the stationary right eigenvector $\mathbf{1} = T\mathbf{1} = (T_1^{(0)} \otimes T_2^{|0} + T_1^{(1)} \otimes T_2^{|1})\mathbf{1} = \mathbf{1}_1 \otimes \mathbf{1}_2 = [1, 1, 1, 1]^\top$, (which once again can be thought of as integrating latent probability).

Although the stationary right eigenvector $\mathbf{1}$ can be decomposed as a product state for a compositional GHMM[14], this is not generally true of the stationary *left* eigenvector $\boldsymbol{\pi}$. In this case, $\boldsymbol{\pi}$ is only a product state if p_2/q_2 is independent of $x^{(1)}$, which would then yield the same local stationary distribution $\boldsymbol{\pi}_2$ for both $T_2^{|0}$ and $T_2^{|1}$, and would imply $\boldsymbol{\pi} = \boldsymbol{\pi}_1 \otimes \boldsymbol{\pi}_2$. Else, if p_2/q_2 depends on $x^{(1)}$, then the joint stationary distribution $\boldsymbol{\pi}$ will be *off of the product-state manifold*, corresponding to mutual information between the two subsystems.

In any case, if we're designing the stochastic process, we could choose for the initial joint distribution to be some non-stationary product state, like $\boldsymbol{\eta}^{(\varnothing)} = [1/2, 1/2]^{\otimes 2}$. Upon subsequent observations, the predictive state would then remain on the product-state manifold—so either the joint or more concise factored representation would suffice for the best possible predictions, as indicated in Fig. 6.

B.2. Forecasting correlated factors requires a joint representation space

If you pick a predictive vector uniformly from the 3-simplex, you *will* obtain a predictive vector that is off of the product-state manifold. For our example of two SNS subsystems, this predictive vector corresponds to a joint distribution over latent-state pairs encoding nonzero mutual information between the two subsystems. In other words, the product states are a measure-zero set within the broader set of joint states. From this perspective, it is quite special for predictive vectors to be on the product-state manifold. Indeed, general update maps correlate the parts, even if they were initially uncorrelated.

[14]More specifically, the stationary right eigenvector $\mathbf{1}$ can be decomposed as a product state for a compositional GHMM when the *type* of factor is fixed for each n—e.g., always an HMM of a fixed size, or always operating on a quantum subsystem of fixed Hilbert-space dimension—such that $\mathbf{1}_n$ is independent of x.

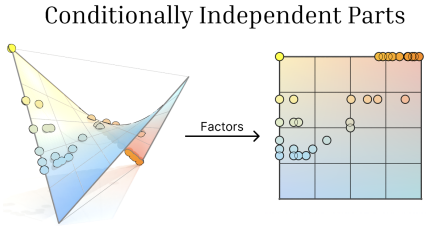

Figure 6. A chain of dependent factors still has a lossless factored representation, since there is a one-to-one linear map from the product-state manifold in the joint space to the Cartesian product of constituent simplices. In contrast to the independent case, the joint distribution over the two 1-simplices can no longer be obtained from the outer-product of the two marginal distributions. (Hence the lack of symmetry of points in the square compared with the independent example.)

So general joint states are relevant. How 'far' are these general joint states from product states? Euclidean distance in the 3-simplex may be a misleading metric. The total correlation may be more relevant. And, for the purpose of predictive loss, relative entropy of next-token predictions is yet more directly relevant.

We will work through these details for the case of two SNS subsystems, as an extension of the independent and conditionally independent example above.

For an arbitrary joint probability $\boldsymbol{\eta} \in \Delta^3$ in the 3-simplex, the closest product state—as measured by relative entropy—is the product of its two marginal distributions. In the language of GHMMs and Sec. A.5, this is the tensor product of the two reduced states. The mutual information between the latent states of the two subsystems is

$$\mathrm{I}[\boldsymbol{\mathcal{S}}_1; \boldsymbol{\mathcal{S}}_2] = \mathrm{D}[\boldsymbol{\eta} \| \boldsymbol{\eta}_1 \otimes \boldsymbol{\eta}_2] \ , \tag{21}$$

where $\boldsymbol{\eta}_1 = \boldsymbol{\eta}\big(I_1 \otimes \mathbf{1}_2\big)$ and $\boldsymbol{\eta}_2 = \boldsymbol{\eta}\big(\mathbf{1}_1 \otimes I_2\big)$.

By standard Euclidean distance, the closest product-state to any joint state is along a line segment normal (i.e., perpendicular) to the local plane of the product-state manifold. However, according to the minimal relative entropy, there is a fixed direction in the 3-simplex from any joint state to its closest product state—the product of its marginals—as shown in Fig. 7.[15]

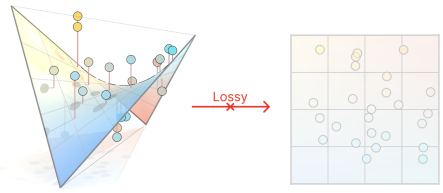

Figure 7. Infinitely many distinct joint states map to the same reduced state on the product-state manifold. These off-manifold joint states encode mutual information between the latent states of the subsystems.

C. Processes used in the experiments

C.1. Factors: Mess3 and Bloch Walk Definition

Here we define the Mess3 and Bloch Walk processes that serve as factors throughout the paper.

C.1.1. MESS3 PROCESS

The Mess3 process (Marzen & Crutchfield, 2017; Shai et al., 2024; Piotrowski et al., 2025) has three hidden states and three observable tokens $\mathcal{X} = \{0, 1, 2\}$. Since Mess3 is a 3-state HMM, it's local predictive vectors live in a 2-simplex. This 2-simplex corresponds to the set of probability distributions over the three hidden states.

The Mess3 process is defined by two parameters, α and x, with dependent quantities $\beta = (1-\alpha)/2$ and $y = 1 - 2x$.

[15] The direction from a joint state to its product of marginals in the 3-simplex is independent of the joint state and proportional to $[1, -1, -1, 1]^\top$. However, in higher dimensions, where there are more degrees of freedom, the direction depends on the joint state.

The labeled transition matrices are:

$$T^{(0)} = \begin{bmatrix} \alpha y & \beta x & \beta x \\ \alpha x & \beta y & \beta x \\ \alpha x & \beta x & \beta y \end{bmatrix} \tag{22}$$

$$T^{(1)} = \begin{bmatrix} \beta y & \alpha x & \beta x \\ \beta x & \alpha y & \beta x \\ \beta x & \alpha x & \beta y \end{bmatrix} \tag{23}$$

$$T^{(2)} = \begin{bmatrix} \beta y & \beta x & \alpha x \\ \beta x & \beta y & \alpha x \\ \beta x & \beta x & \alpha y \end{bmatrix} . \tag{24}$$

The net transition matrix

$$T = T^{(0)} + T^{(1)} + T^{(2)} = (\alpha + 2\beta) \begin{bmatrix} 1-2x & x & x \\ x & 1-2x & x \\ x & x & 1-2x \end{bmatrix} \tag{25}$$

has a uniform stationary distribution $\boldsymbol{\pi} = \boldsymbol{\pi} T = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$, and a stationary right eigenstate of $\mathbf{1} = T\mathbf{1} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^\top$.

C.1.2. Bloch Walk process

Bloch Walk (Riechers et al., 2025) has no finite HMM representation but can be simulated via interactions with a single qubit of quantum memory. It thus has a simple GHMM and its predictive vectors all lie on the x-z slice of the Bloch ball.

In general, valid states of a qubit can be described by the extended Bloch vector $\vec{b} = [1, b_x, b_y, b_z]$. Interactions with a qubit induce linear transformations on this Bloch vector—and projective transformations, once the state is properly renormalized.

The Bloch Walk process is parametrized by $\alpha > 0$ and $\beta \in \mathbb{R}$. Let $\gamma = 1/(2\sqrt{\alpha^2 + \beta^2})$.

The minimal three-dimensional GHMM for this Bloch Walk process is obtained by projecting out the non-utilized y-component of the Bloch ball:

$$T^{(0)} = \begin{bmatrix} 1/4 & 0 & 2\alpha\beta\gamma^2 \\ 0 & (\alpha^2-\beta^2)\gamma^2 & 0 \\ 2\alpha\beta\gamma^2 & 0 & 1/4 \end{bmatrix} \tag{26}$$

$$T^{(1)} = \begin{bmatrix} 1/4 & 0 & -2\alpha\beta\gamma^2 \\ 0 & (\alpha^2-\beta^2)\gamma^2 & 0 \\ -2\alpha\beta\gamma^2 & 0 & 1/4 \end{bmatrix} \tag{27}$$

$$T^{(2)} = \begin{bmatrix} 1/4 & 2\alpha\beta\gamma^2 & 0 \\ 2\alpha\beta\gamma^2 & 1/4 & 0 \\ 0 & 0 & (\alpha^2-\beta^2)\gamma^2 \end{bmatrix} , \text{ and} \tag{28}$$

$$T^{(3)} = \begin{bmatrix} 1/4 & -2\alpha\beta\gamma^2 & 0 \\ -2\alpha\beta\gamma^2 & 1/4 & 0 \\ 0 & 0 & (\alpha^2-\beta^2)\gamma^2 \end{bmatrix} , \tag{29}$$

which can be interpreted as acting from the right on the coefficients $[c, b_x, b_z]$ of the ordered operator basis $(I/2,\, \sigma_x/2,\, \sigma_z/2)$, where σ_x and σ_z are two of the familiar Pauli matrices from quantum mechanics. See Riechers et al. (2025) for further details of this process.

The net transition operator

$$T = \sum_{x \in \mathcal{X}} T^{(x)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 + 2(\alpha^2-\beta^2)\gamma^2 & 0 \\ 0 & 0 & 1/2 + 2(\alpha^2-\beta^2)\gamma^2 \end{bmatrix} \tag{30}$$

has the stationary left eigenvector $\boldsymbol{\pi} = \boldsymbol{\pi} T = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$ and stationary right eigenvector $\mathbf{1} = T\mathbf{1} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^\top$.